



Module Objective: In previous weeks, your Drow Ranger agent used A* Search to find a physical path to the goal. However, it blindly assumed every non-wall tile was safe. Today, we implement a **Knowledge Base (KB)** and a **Forward Chaining Inference Engine**. Your agent must now mathematically prove a cell is logically *Feasible* before taking a step, using real-time rules about map vision and enemy threats.

Part 1: The Logic Engine Architecture

Step 1.1: Building the Knowledge Base (KB)

Instead of hardcoding hundreds of nested `if-else` statements in our agent's movement logic, we will build a declarative Knowledge Base. You need to design a Python class that stores **Facts** (current percepts) and **Rules** (Horn Clauses).

1. Create a new file named `logic_engine.py`.
2. Implement a `KnowledgeBase` class based on the following structural requirements:

```
CLASS KnowledgeBase:
    // Attributes
    DECLARE facts AS Set                // To store unique st
    DECLARE rules AS List               // To store rules as

    // Methods
    FUNCTION tell_fact(fact_string):
        ADD fact_string TO facts

    FUNCTION tell_rule(premise_list, conclusion_string):
        APPEND Tuple(premise_list, conclusion_string) TO rules
```

```
FUNCTION clear_facts():  
    EMPTY the facts Set
```

Part 2: Implementing Forward Chaining

Step 2.1: The Inference Algorithm

The agent needs an Inference Engine to deduce new information from its raw sensors. Translate the following Data-Driven **Forward Chaining** algorithm into a Python method inside your `KnowledgeBase` class.

1. Write the `forward_chain()` method.
2. Ensure you use a `while` loop that continues to iterate over the rules until a full pass results in absolutely no new facts being deduced.

```
FUNCTION forward_chain():  
    DECLARE new_facts_added = TRUE  
  
    WHILE new_facts_added == TRUE:  
        new_facts_added = FALSE  
  
        FOR EACH (premises, conclusion) IN rules:  
            IF conclusion IS NOT IN facts:  
  
                // Modus Ponens Check  
                IF ALL premises ARE IN facts:  
                    ADD conclusion TO facts  
                    new_facts_added = TRUE
```

Part 3: Hooking Logic into the Grid Game

Step 3.1: Defining the Game Constraints

Now we translate the game's safety constraints into our logic engine using Propositional Logic.

1. In your main `agent.py`, instantiate the KB in the `__init__` method:
`self.kb = KnowledgeBase()`
2. Use your `tell_rule` method to define the following safety rules:
Rule 1: `TargetVisible` $\wedge$ `HasDust` $\Rightarrow$ `SafeToEngage`
Rule 2: `SafeToEngage` $\wedge$ `BloodseekerMissing` $\Rightarrow$ `Retreat`

Step 3.2: Validating Feasibility in A*

Modify your existing A* pathfinding algorithm to consult the Knowledge Base before expanding a node.

1. Locate the neighbor-generation loop in your A* code (where you normally check for physical walls).
2. Before adding a neighbor to the `open_list`, clear the KB facts.
3. Feed the current percepts for that specific tile into the KB using `tell_fact()`.
4. Run `self.kb.forward_chain()`.
5. If `'Retreat'` is deduced and exists in `self.kb.facts`, mark the tile as **Infeasible**. Skip it (do not add it to the open list), even if it is physically reachable.

Part 4: Self-Evaluation Test Cases

4.1 Automated Validation

Create a file named `test_logic.py`. Copy and run the following Python assert statements to verify your Forward Chaining engine works correctly mathematically before you attempt to integrate it into the visual grid.

```
from logic_engine import KnowledgeBase

def test_forward_chaining():
    kb = KnowledgeBase()

    # Add Domain Rules
    kb.tell_rule(['TargetVisible', 'HasDust'], 'SafeToEngage')
    kb.tell_rule(['SafeToEngage', 'BloodseekerMissing'], 'Retreat')
```

```

# Test Case 1: Safe Engagement
kb.clear_facts()
kb.tell_fact('TargetVisible')
kb.tell_fact('HasDust')
kb.forward_chain()
assert 'SafeToEngage' in kb.facts, "Test 1 Failed: Should deduce SafeToEngage"
assert 'Retreat' not in kb.facts, "Test 1 Failed: Should Not deduce Retreat"

# Test Case 2: Unsafe Engagement (Bloodseeker Missing)
kb.clear_facts()
kb.tell_fact('TargetVisible')
kb.tell_fact('HasDust')
kb.tell_fact('BloodseekerMissing')
kb.forward_chain()
assert 'Retreat' in kb.facts, "Test 2 Failed: Should deduce Retreat"

print("✅ All Logic Engine Test Cases Passed!")

if __name__ == "__main__":
    . . .

```

Part 5: Theory-to-Implementation Mapping

Based on the code you just wrote and the lecture on Logical Reasoning, complete the following analytical questions to explain *how* your code implements the theory.

1. Reachability vs. Feasibility (Apply)

In Step 3.2, you modified the A* algorithm's neighbor-generation loop. How does your new code differentiate between checking if a node is "Reachable" (like a wall collision) versus checking if it is "Feasible"? Explain how the Knowledge Base enforces feasibility.

Reachability checks whether a tile can physically be reached by the agent. A* checks this by making sure the tile is inside the grid, is not a wall, and has not already been visited.

Feasibility checks whether the tile is logically safe. Before adding a neighbor to the A* frontier, the agent clears the Knowledge Base, adds the facts for that tile, and runs `forward_chain()`. If `Retreat` is derived, the tile is considered infeasible and is skipped.

Therefore, reachability is based on the physical environment, while feasibility is based on the logical rules in the Knowledge Base.

2. Declarative vs. Procedural Paradigms (Analyze)

Why did we build a `KnowledgeBase` class with a `forward_chain()` loop instead of simply writing `if 'TargetVisible' in percepts and 'HasDust' in percepts: directly inside the agent's movement code? What happens to the agent's code complexity if we add 50 new game rules?`

We use a Knowledge Base because it separates the rules from the agent's movement code.

Instead of writing conditions such as `if TargetVisible and HasDust` directly inside the agent, we store the rules in the Knowledge Base and use `forward_chain()` to apply them.

This makes the system easier to maintain and extend. If new game rules are added, we can add them to the Knowledge Base without changing the main A* movement logic.

3. Modus Ponens & Horn Clauses (Understand)

The rule `['TargetVisible', 'HasDust'] ⇒ 'SafeToEngage'` is formally known as a Horn Clause. Briefly explain how your Python implementation of the `forward_chain()` while loop acts as a mathematical execution of the *Modus Ponens* inference rule.

The rule: $\text{TargetVisible} \wedge \text{HasDust} \rightarrow \text{SafeToEngage}$

is a Horn Clause because it has multiple premises leading to one conclusion.

Our `forward_chain()` method applies Modus Ponens by checking whether all the premises of a rule are present in the Knowledge Base. If they are, the conclusion is added as a new fact.

For example:

$\text{TargetVisible} \wedge \text{HasDust} \rightarrow \text{SafeToEngage}$

The process continues until no new facts can be derived. This also allows the newly derived `SafeToEngage` fact to be used to derive `Retreat` when `BloodseekerMissing` is present.

Finalize and Lock Lab 05 Implementation

✅ **Lab 05 Theory-to-Implementation Mapping completed!**



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